

Athena

Deep Research Report Generator

Framework: OpenAI Agents SDK + Streamlit | Version: 1.0 | Date: June 2026
Repository: bibhu2020/agenticaï (Hugging Face Space)

1. Overview

Athena is an AI-powered deep research platform. Given a research query and parameters (depth, report format), it plans a set of targeted web searches, executes them in parallel, then writes a comprehensive report and optionally sends it by e-mail. Reports can be exported as PDF. The app uses the OpenAI Agents SDK with an MCP-based web search server.

2. System Design

2.1 Components

Component	Technology	Purpose
app.py	Streamlit	UI — query input, depth selector, live status stream, PDF export
Orchestrator	Python class	Coordinates the four-agent pipeline; yields status strings to UI
Planner Agent	OpenAI Agents SDK	Decomposes query into N targeted search sub-queries
Search Agent	OpenAI Agents SDK	Executes each sub-query via MCP web-search; returns snippets
Writer Agent	OpenAI Agents SDK	Synthesises all search results into a formatted markdown report
Email Agent	OpenAI Agents SDK	Sends the finished report to a recipient via email tool
mcp-web-search server	MCPServerStdio	Standalone MCP server providing web_search tool
model_factory.py / llm.py	openai.AsyncOpenAI	LLM factory; Google Gemini 2.5 Flash default

Component	Technology	Purpose
Guardrail	InputGuardrail	Blocks non-research queries before planning phase

2.2 Architecture Diagram



3. Agentic Flow

Phase	Agent	Input	Output
Plan	Planner Agent	User query + num_searches	WebSearchPlan: list of search sub-queries
Search	Search Agent (×N, parallel)	Each sub-query	WebSearchItem: title, url, snippet
Write	Writer Agent	All WebSearchItems + format hint	ReportData: markdown_report string

Phase	Agent	Input	Output
Email (opt.)	Email Agent	Report + recipient address	Sent confirmation

4. Depth Levels

Level	Searches	Typical Use
Quick	5	Fast summaries, fact checks
Standard	10	Balanced research for most topics
Deep	15	Comprehensive academic-style reports

5. Configuration

Env Var	Required for
OPENAI_API_KEY	Default LLM
GOOGLE_API_KEY	Gemini model
SERPER_API_KEY	Web search MCP server
LANGFUSE_SECRET_KEY / PUBLIC_KEY	Optional distributed tracing